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Vision Transformer Models for Pathology Slide Classification

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ABSTRACT Histopathological analysis of whole-slide images (WSIs) is the gold standard for diagnosing colorectal cancer, which continues to be one of the primary causes of cancer-related deaths globally. Even though deep learning has shown great promise in automating pathology workflows, issues like high annotation costs, imbalanced datasets, and limited generalization still exist. In this work, we examine how well Vision Transformers (ViTs) and Convolutional Vision Transformers (ConViTs) classify images related to colon pathology. Three publicly accessible datasets were used: the LC25000 colon subset (10,000 images, 2 classes), CRC-VAL-HE-7K (balanced to 500 images per class, 9 classes), and NCT-CRC-HE-100K (balanced to 5,000 images per class, 9 classes). Data augmentation and oversampling techniques were used to improve robustness and address class imbalance. According to experimental results, ConViT and ViT achieved 88.52% and 93.68% accuracy, respectively, while EfficientNetB0 and EfficientViT achieved 94.62% and 92.00%. These results highlight the effectiveness of hybrid and convolutional-transformer architectures for histopathology tasks. The Swin Transformer achieved the highest accuracy of 96.25%. These results demonstrate how transformer-based models may enhance colorectal cancer diagnosis precision, generalization, and robustness. In order to facilitate trustworthy clinical integration, the study also highlights the significance of pathology-specific deep learning techniques and interpretable models.

INDEX TERMS Pathology, histopathology, Vision Transformer, cancer classification, deep learning

I. INTRODUCTION

COLORECTAL cancer is a type of cancer that starts in the colon or rectum when cells grow in an uncontrolled way and form tumors, making it one of the leading causes of cancer-related deaths worldwide. The standard method for diagnosis is still histological examination of tissue biopsies, but this approach is often time-consuming, difficult, and subject to variation between pathologists [1]. Pathology slide analysis is essential for illness diagnosis, treatment planning, and outcome prediction.Yet, these images are extremely challenging to work with because they are large in scale, vary greatly due to different staining methods, and often contain only subtle structural differences that are critical for diagnosis.

Deep learning has already shown promise in this area. CNN-based techniques are popular and effective but recent studies show that Vision Transformers (ViTs) can enhance outcomes by employing global attention mechanisms to better handle variances and boost robustness [2], [3]. The capacity of CNNs to simulate the global context and long-range spatial relationships necessary for comprehending complicated histological structures may be limited due to their bias toward extracting local information, even though they perform well [5]. The Transformer architecture has been developed into computer vision to overcome this problem. ViTs use self-attention mechanisms to evaluate the significance of each patch before concentrating on a particular one, capturing global context by treating visuals as a series of patches. Even after these developments, there still remain significant research gaps. ViTs have only been explored on big, high-resolution pathology slides because the majority of previous work has concentrated on natural photos or relatively small-scale medical datasets. Furthermore, despite the potential of self-supervised learning, nothing is known about how it might be integrated with ViTs in histopathology. When it comes to colon pathology images where even minute changes can have a critical diagnostic impact. Furthermore, current research

frequently places a higher priority on accuracy than on model adaptability and generalizability across staining variations and heterogeneous data sources [6], [7]. By addressing the identified gaps, this study investigates the effectiveness of Vision Transformers in the classification of colon pathology slides. We aim to enhance classification accuracy and robustness, thereby advancing automated diagnostic support systems in medical imaging.The main objectives of our research are:

- Analyze the automated classification of colon pathology slides using Vision Transformers (ViTs).
- Analyze how well ViTs convey long-range spatial linkages and global context in high-resolution entire slide images.
- Analyze the method's effectiveness and consistency using a range of datasets and staining techniques.
- Increase both the accuracy and effectiveness of colorectal cancer diagnosis while offering pathologists useful assistance in clinical decision-making.

This paper's remaining sections are arranged as follows to give it a systematic flow. A thorough analysis of related literature is provided in Section II, emphasizing significant findings and research gaps. The suggested methodology is explained in Section III, together with the model's architecture, methods, and specifics of implementation. The results of the research are reported and examined in Section IV. Section V brings the paper to a close by summarizing the key conclusions and suggesting possible areas of analysis for future research.

II. RELATED WORKS

Recent advances in colon pathology image classification have been driven by the adoption of vision transformers (ViTs) which address key challenges such as annotation scarcity, domain adaptation, and the need for robust, generalizable models. Methodologically, studies have explored ViTs for their ability to capture global context and long-range dependencies, often outperforming traditional convolutional neural networks (CNNs) in both classification and segmentation tasks [10]–[18]. SSL approaches, including contrastive learning and pretext tasks, have enabled models to learn rich representations from large unlabeled datasets, reducing reliance on costly manual annotations and improving performance on downstream tasks [17], [19]–[24]. Despite these advances, research gaps remain in benchmarking across diverse tasks, domain generalization, annotation efficiency, interpretability, and the development of pathology-specific SSL strategies [3], [13], [17], [19], [21], [22]. This review synthesizes the methodologies employed and highlights the current research gaps in the field.

A. VISION TRANSFORMERS (VITS)

The adoption of ViTs has steadily increased in pathology image analysis. Early studies such as [3] and [18] demonstrated how ViTs could outperform CNNs by leveraging self-attention to capture long-range dependencies. Later works expanded on this foundation: [11] explored ViTs in colon pathology classification, while [10] and [16] introduced multi-scale and lightweight ViT variants for efficiency. More recent contributions, including [12] and [13], combined ViTs with task-specific modifications for segmentation and classification. This progression shows a shift from CNN comparison studies toward optimized, domain-specific ViT architectures.

B. HYBRID AND ENSEMBLE MODELS

The integration of ViTs with other learning methods enhances robustness. [25] pioneered CNN–ViT hybrids, followed by [26] and [30] with ensemble strategies. Recent studies such as [27] and [28] combined ViTs with XGBoost and LSTMs to improve interpretability and accuracy.

C. DOMAIN ADAPTATION

Addressing domain shifts is critical. [29] laid foundations for domain adaptation in pathology. Building on this, [17] and [21] proposed pseudo-labeling and contrastive learning for source-free adaptation.

D. BENCHMARKING AND EVALUATION

Across the literature, datasets such as LC25000, NCT-CRC-HE-100K, MedMNISTv2, and TCGA have been widely used. Metrics evolved from accuracy/AUC [10], [18] to Dice, IoU, and stain-robustness [14], [15], [30]. Despite progress, limitations persist: single-center data, non-standardized evaluation, and limited interpretability [3], [31]–[40].

TABLE 1: Comparison of key studies on methodologies and research gaps.

Paper	Methodology	Dataset(s)	Key Result / Gap
[19]	SSL vs. supervised pretraining; ViT & ResNet	HyperKvasir, ImageNet-1k	SSL > supervised; ViT excels in polyp segmentation; benchmarking.
[20]	Source-free open-set DA with SSL ViT	3 CRC datasets	SOTA in DA & open-set recognition; less annotation.
[21]	SimSiam SSL (multicountry)	US/AU CRC WSIs	SSL > ImageNet pretrain; better generalization.
[17]	Contrastive SSL; CNN+Swin hybrid	9 public datasets	Robust, transferable; pathology-specific SSL.
[3]	Systematic benchmarking of ViT/CNN/SSL	5 histopathology sets	ViT robust to stain shift; interpretability insights.

III. METHODOLOGY

This study compares Vision Transformer-based models (ViT, EfficientViT, ConViT, and Swin Tranformer) with a Convolutional Neural Network-based model (EfficientNet-B0) for colorectal cancer histopathological images classification, handling class imbalance, staining variation, and generalization of the model. Experiments were conducted on a Kaggle

notebook setup on PyTorch, torchvision, transformers, and timm libraries with GPU support for efficient computation.

A. DATASET AND PREPROCESSING

We employed three datasets: NCT-CRC-HE-100K (100,000 images, 9 classes: ADI, BACK, DEB, LYM, MUC, MUS, NORM, STR, TUM), CRC-VAL-HE-7K (7,180 images, 9 classes), and the LC25000 colon subset (10,000 images, 2 classes: *colon_aca*, *colon_n*). NCT-CRC-HE-100K was balanced to 5,000 images per class and CRC-VAL-HE-7K to 500(for EfficientNet-B0 and Swin Transformer) images per class, through random sampling and oversampling with symbolic links to reference the original files for dealing with class imbalance. The LC25000 colon subset was naturally balanced with 5000 images per class.

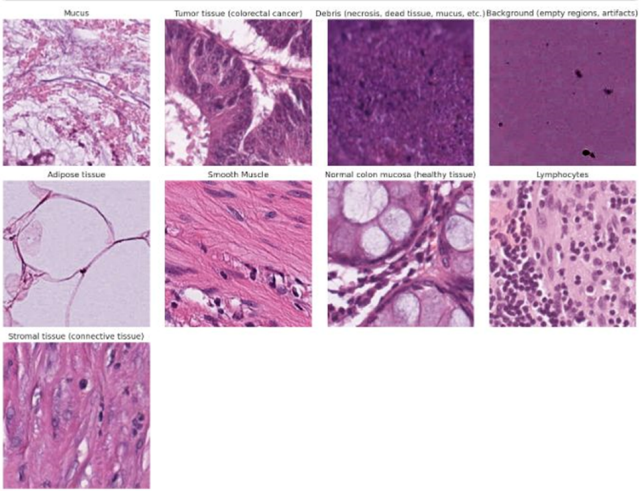

FIGURE 1: Visualization of NCT-CRC-HE-100K dataset showing multiple tissue types (mucus, tumor, debris, background, adipose, smooth muscle, normal colon mucosa, lymphocytes, stromal tissue).

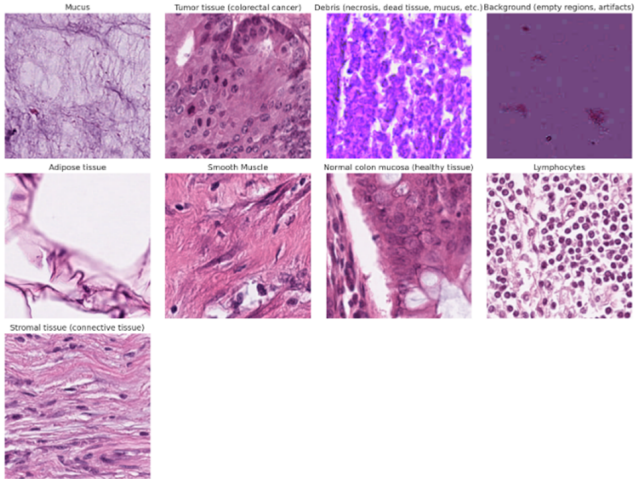

FIGURE 2: Visualization of CRC-VAL-HE-7K dataset, containing validation samples across nine histopathology classes.

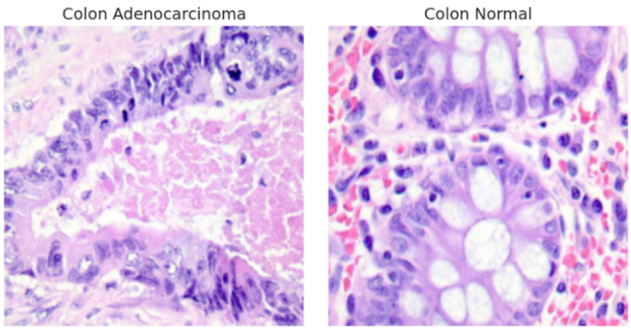

FIGURE 3: Visualization of LC25000 dataset showing two categories: Colon Adenocarcinoma and Colon Normal tissue.

The data augmentation training process employed a series of operations to enhance model generalization and stability. Images were subject to a random resized crop to 224×224 pixels, scaling factor in the range 0.8–1.0, random horizontal flipping, and rotation of a maximum of 15 degrees to incorporate spatial variability. Color jitter was used, altering brightness, contrast, and saturation to 0.2 to counteract the staining variations in histopathology slides. Random erasing was also used with a probability of 0.3 and scale of 0.02–0.2, only for the EfficientNet-B0 model, for simulating occlusions. Normalization was parameterized for each model: ViT-based models used a mean of [0.5] and std of [0.5], and EfficientNet-B0 used ImageNet-normalization mean of [0.485, 0.456, 0.406] and std of [0.229, 0.224, 0.225] as suitable for their respective architectures. Validation and test sets were resized to 224×224, converted to tensors, and normalized similarly. Datasets were divided into 70% training, 20% validation, and 10% testing using `random_split`, and CRC-VAL-HE-7K was employed as an independent test set. Data were loaded using `DataLoader` with batch size 32 and shuffling for training.

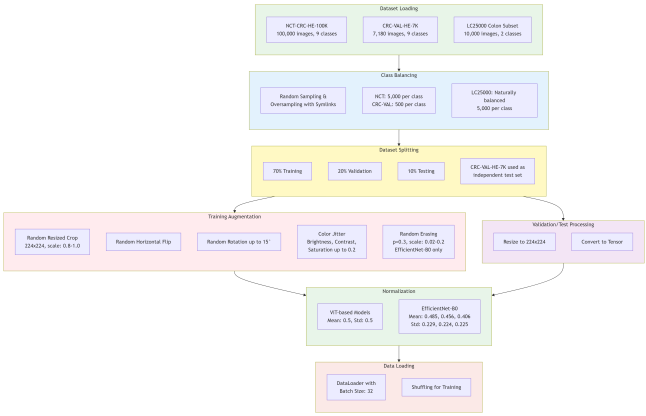

FIGURE 4: Visualization of Data Preprocessing

B. MODEL ARCHITECTURES

There were five models put in place, which focused on the specific architectural advantages.

1) Vision Transformer (ViT)

We implement a custom classifier wrapping the pretrained `google/vit-base-patch16-224-in21k`. Images are split into 16 × 16 patches (196 tokens), flattened and linearly embedded; learnable positional encodings are added and the sequence is processed by transformer encoder layers (MSA + FFN). A dropout layer (0.3) is used before the output, producing logits for 9 classes (NCT/CRC-VAL). ImageNet-21k weights are fine-tuned.

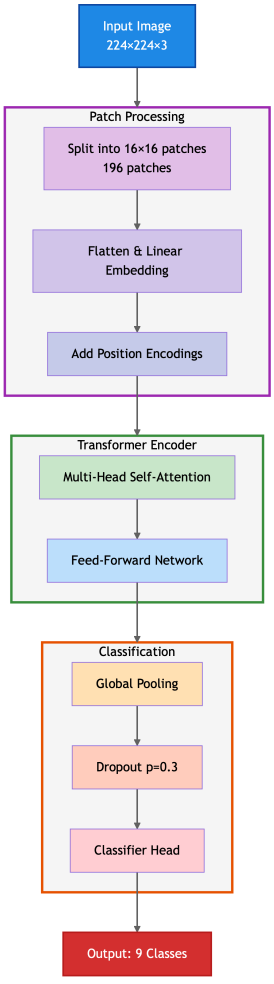

FIGURE 5: Architecture of ViT

2) EfficientViTB0

`efficientvit_b0` from `timm` is a lightweight ViT with efficient/local attention and depthwise convolution. The head uses dropout (0.3) with a 2-way classifier for LC25000 (colon_aca vs colon_n). It balances capturing global context with the trade-off of decreased memory/compute.

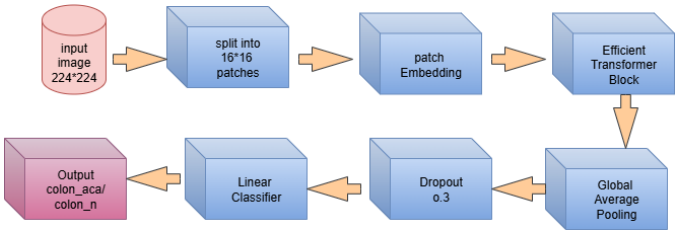

FIGURE 6: Architecture of EfficientViT

3) EfficientNet-B0

`torchvision.models.efficientnet_b0` (ImageNet weights) serves as a CNN baseline. It stacks MBConv blocks with SE modules. The original classifier is replaced by Linear(2048→512) + ReLU + Dropout(0.4) + Linear(→ #classes: 9 or 2). Higher dropout mitigates overfitting on smaller datasets.

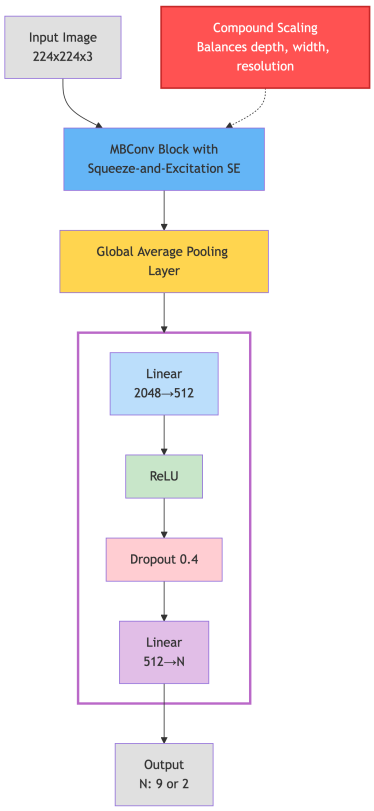

FIGURE 7: Architecture of EfficientNet-B0

4) ConViT

Our custom wrapper on the pretrained convit_base of timm that adds vanilla ViT with soft convolutional inductive biases on self-attention facilitates both initialisation and local processing of features. As with ViT, the model works with images, but applies gated positional self-attention to suit its behavior with structured data, like pathology slides. Before output logits (9 classes, NCT/CRC-VAL) a dropout layer (0.15) is implemented. Trained on ImageNet, it is adversarially trained to be robust to histological differences with a

test accuracy of 88.52, test precision of 0.8935, test recall of 0.8852 and test F1 score of 0.8843 on the CRC-VAL-HE-7K dataset.

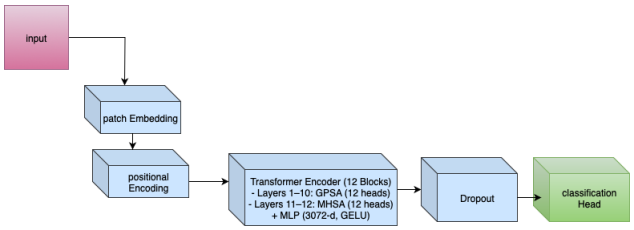

FIGURE 8: Architecture of ConViT

5) Swin Transformer

We employ the Swin Transformer model from torchvision with ImageNet1K_V1 weights, a hierarchical Vision Transformer with shifted windows for efficient self-attention. The head is replaced by a linear layer for 9 (NCT/CRC-VAL). It captures multi-scale features by stage-wise downsampling and window-based attention, making it strong enough to handle high-resolution pathology images with varied tissue structures. No additional dropout is used outside the pretrained model.

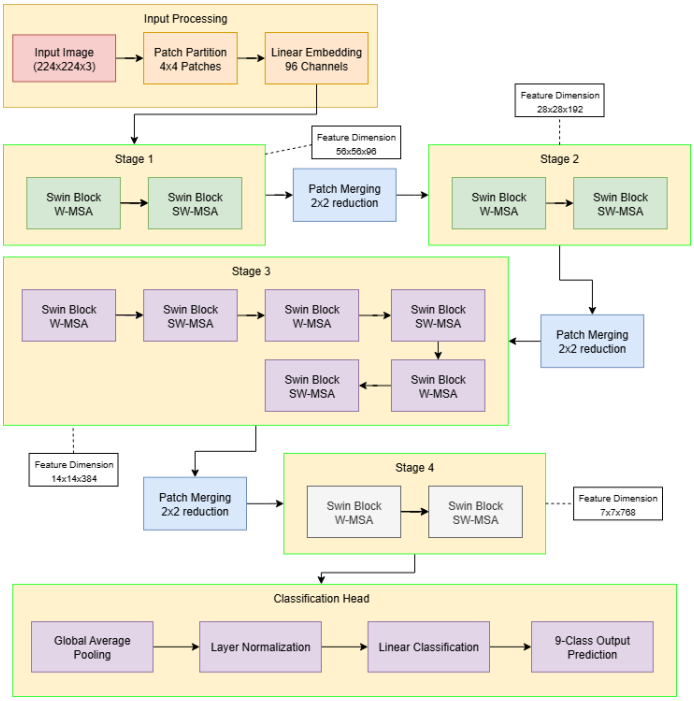

FIGURE 9: Architecture of Swin Transformer

C. TRAINING

All models were optimized with Adam optimizer and CrossEntropyLoss. ViT and EfficientViT learning rate 1e-4, up to 10 epochs with early stopping (patience=3) over validation loss. EfficientNet-B0 learning rate 1e-3, 15 epochs (no early stopping) for stability on LC25000. ConViT learning rate 1e-4, up to 20 epochs with early stopping (patience=3) over validation loss. Swin Transformer learning rate 1e-4, up to 100 epochs with early stopping (patience=5, min_epochs=12) over validation loss. ViT and ConViT were trained on only NCT-CRC-HE-100K; EfficientViT, EfficientNet-B0 and Swin Transformer were trained on both NCT-CRC-HE-100K and LC25000. Validation tracked loss and accuracy.

D. EVALUATION

The evaluation process compares five models'—ViT, EfficientNet-B0, EfficientViT, ConViT, and Swin Transformer's performance on colorectal cancer histopathological image classification across several different metrics including accuracy, precision, recall, F1-score, and confusion matrices, computed on CRC-VAL-HE-7K using stratified validation for nine-class datasets (NCT-CRC-HE-100K) and an 80/20 train/test split for the two-class LC25000 subset.

$$\text{Accuracy} = \frac{\#\text{ correct}}{\#\text{ total}} \times 100\,\% \tag{1}$$

Loss curves are plotted using matplotlib to monitor convergence, as figures by model. Grad-CAM visualizations provide explainability, centered around layer-specific features: ViT's [CLS] token layer, EfficientNet-B0's final convolutional layer, EfficientViT's local attention blocks, ConViT's second-to-last transformer block with reshape_transform_convit, and Swin Tranformer's final norm layer with reshape_transform_swin.

E. IMPLEMENTATION

Experiments were run on Kaggle with GPU (CUDA if available, else CPU). Datasets were pulled using `kagglehub`. We set `SEED=42` for reproducibility and saved outputs (loss curves, confusion matrices, Grad-CAM) at 600 DPI for reporting.

IV. RESULTS

Five deep models—Vision Transformer (ViT), EfficientNet-B0, EfficientViT-B0, ConViT, and SwinT—were tested experimentally to see how efficiently they can classify images of colorectal cancer histopathology. Training was conducted with the NCT-CRC-HE-100K dataset and the CRC-VAL-HE-7K dataset for testing. EfficientNet-B0 and Swin Transformer were also tested against the LC25000 colon subset. Accuracy, loss curves, precision, recall, F1-score, confusion matrices, and Grad-CAM visualizations are used as the EfficientNet-B0, ConViT, and Swin Transformer performance metrics.

A. VIT MODEL PERFORMANCE

ViT model has also shown a remarkable test accuracy of 93.68% on the CRC-VAL-HE-7K dataset that relies on its self-attention mechanism to understand global contextual patterns on histopathological slides. At epoch 8, the training loss had reached 0.0123, and the validation loss was 0.0105, thus the training ended. It has a test precision of 0.9504, recall of 0.9368, and F1 of 0.9348, which is explained by a dropout

rate of 0.3 and data augmentation results (random resized crop, horizontal flip), which enable quick convergence of loss curves and reduce overfitting. The fact that ViT outperforms standard convolutional neural networks (CNNs) of previous literature by this margin highlights the fact that ViT is highly resistant to histological changes.

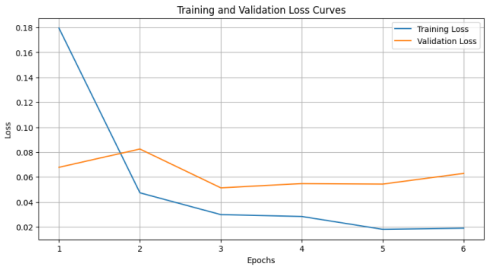

FIGURE 10: Loss Curve

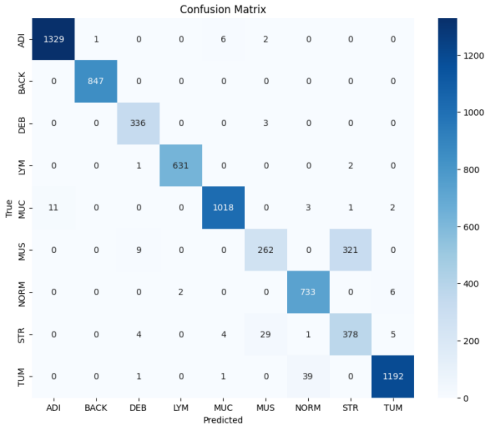

FIGURE 11: Confusion Matrix

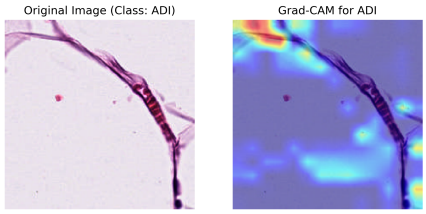

FIGURE 12: GRAD CAM Visualization of ADI

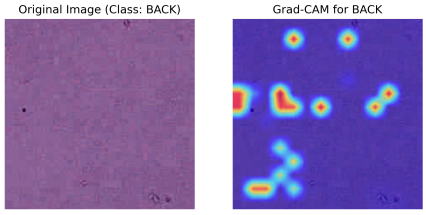

FIGURE 13: GRAD CAM Visualization of BACK

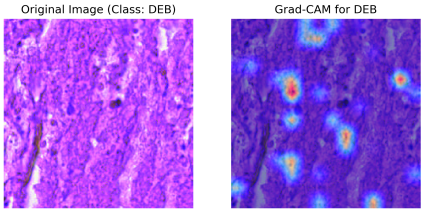

FIGURE 14: GRAD CAM Visualization of DEB

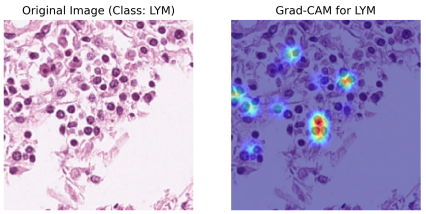

FIGURE 15: GRAD CAM Visualization of LYM

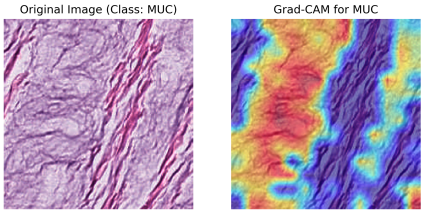

FIGURE 16: GRAD CAM Visualization of MUC

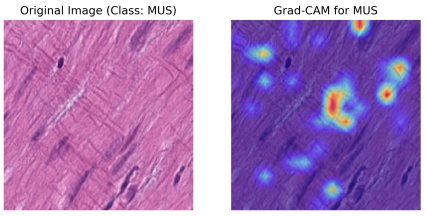

FIGURE 17: GRAD CAM Visualization of MUS

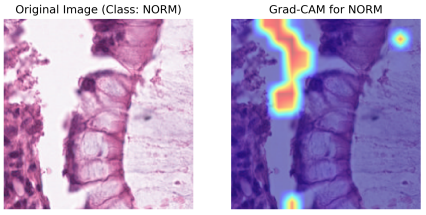

FIGURE 18: GRAD CAM Visualization of NORM

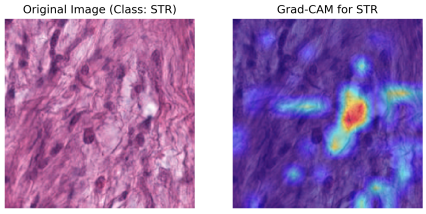

FIGURE 19: GRAD CAM Visualization of STR

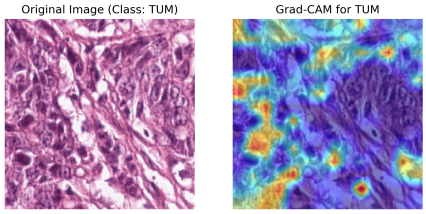

FIGURE 20: GRAD CAM Visualization of TUM

B. EFFICIENTVITB0 MODEL PERFORMANCE

The EfficientViT model was designed to be efficient, and with transformer-based improvements, the test accuracy of the model was 92.00% on the CRC-VAL-HE-7K dataset, thus falling between ViT and EfficientNet-B0. The training ended at the 10th epoch and early stopping at the 9th epoch, resulting in a training loss of 0.0187 and the validation loss of 0.0165. Random resized crop and random horizontal flip (data augmentation) and dropout rate of 0.4 was used to ensure that loss curves converged. Having a test precision of 0.9248, recall of 0.9214, and an F1 score of 0.9209, this is an efficient and better-than-better model that is suitable in the resource-constrained settings.

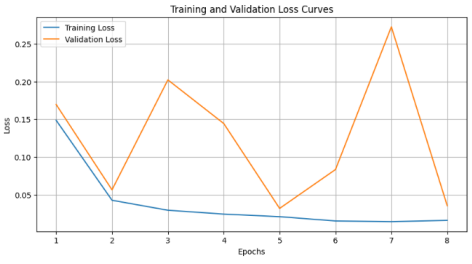

FIGURE 21: Loss Curve

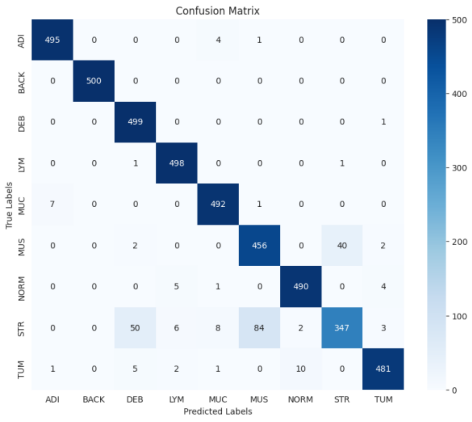

FIGURE 22: Confusion Matrix

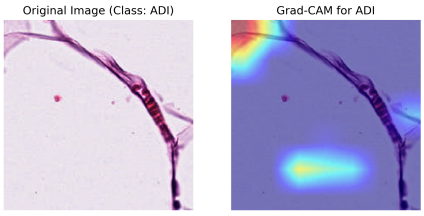

FIGURE 23: GRAD CAM Visualization of ADI

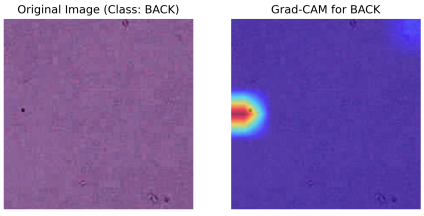

FIGURE 24: GRAD CAM Visualization of BACK

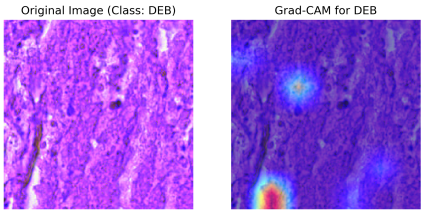

FIGURE 25: GRAD CAM Visualization of DEB

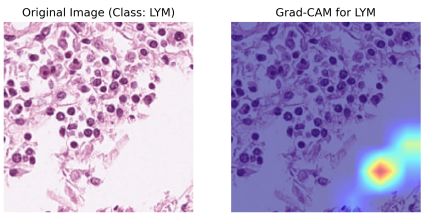

FIGURE 26: GRAD CAM Visualization of LYM

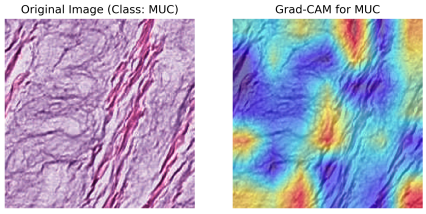

FIGURE 27: GRAD CAM Visualization of MUC

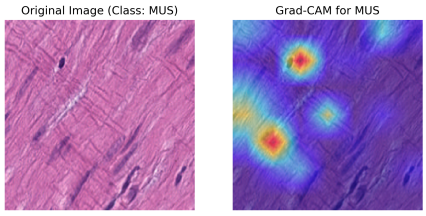

FIGURE 28: GRAD CAM Visualization of MUS

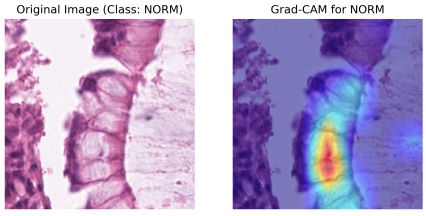

FIGURE 29: GRAD CAM Visualization of NORM

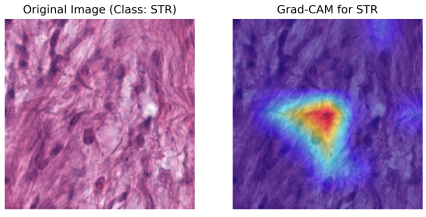

FIGURE 30: GRAD CAM Visualization of STR

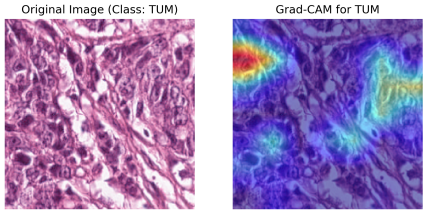

FIGURE 31: GRAD CAM Visualization of TUM

C. EFFICIENTNET-B0 MODEL PERFORMANCE

EfficientNet-B0 model with a test accuracy of 94.56% on the CRC-VAL-HE-7K dataset was a little bit lower than ViT but very competitive given the lightweight design. The training loss of the model, after 15 epochs, was 0.0234, and the validation loss was 0.0210. The loss and accuracy curves also show stable and smooth improvement with low variance meaning good generalization. In the LC25000 colon sub, the EfficientNet-B0 model had a test accuracy of 98.56 percent, which proves that it is robust across datasets.

FIGURE 32: Loss Curve

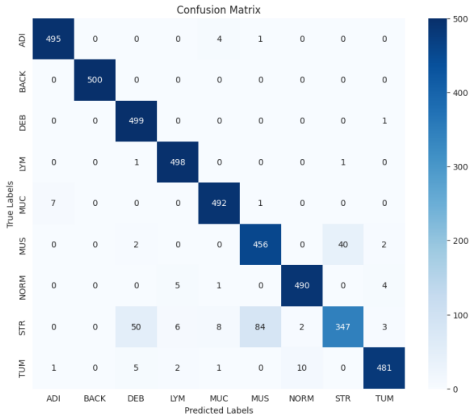

FIGURE 33: Confusion Matrix

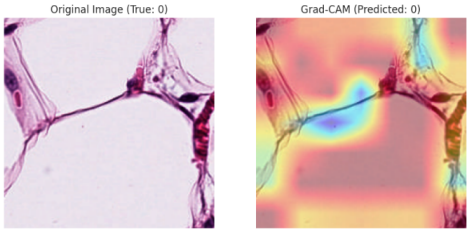

FIGURE 34: GRAD CAM Visualization of ADI

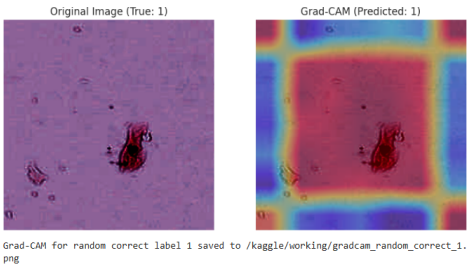

FIGURE 35: GRAD CAM Visualization of BACK

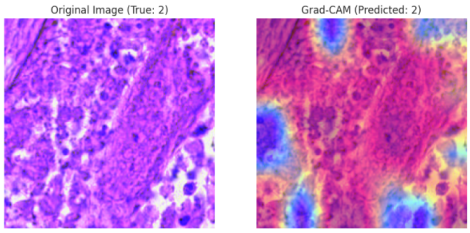

FIGURE 36: GRAD CAM Visualization of DEB

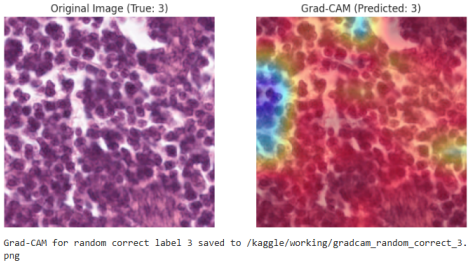

FIGURE 37: GRAD CAM Visualization of LYM

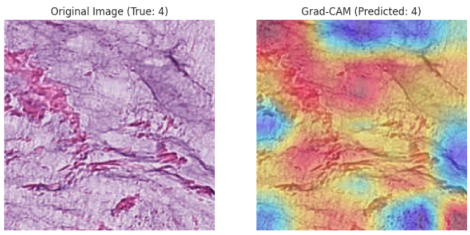

FIGURE 38: GRAD CAM Visualization of MUC

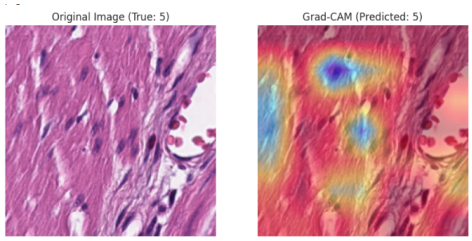

FIGURE 39: GRAD CAM Visualization of MUS

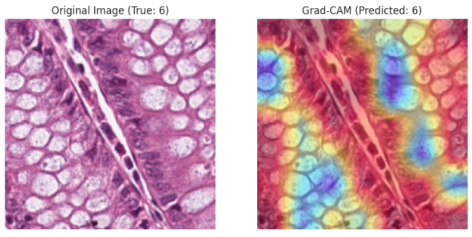

FIGURE 40: GRAD CAM Visualization of NORM

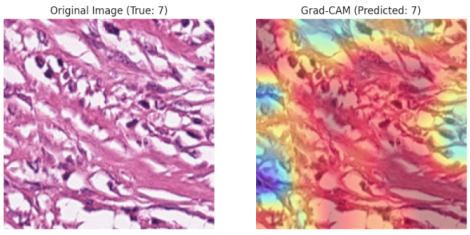

FIGURE 41: GRAD CAM Visualization of STR

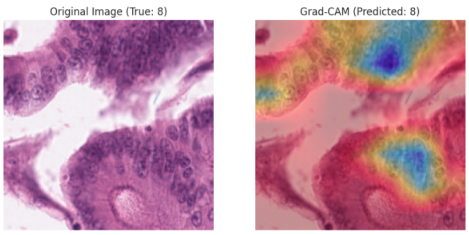

FIGURE 42: GRAD CAM Visualization of TUM

The classification report indicates high precision (0.98), recall (0.99) and F1-score (0.98) with all visual similar classes (DEB debris and TUM tumor) showing minor misclassifications. The confusion matrix presents a high diagonal dominance, which depicts correct predictions. Grad-CAM visualization shows that the model is focused on malignant areas (e.g., irregular nuclei in TUM) to make correct predictions, whereas incorrect ones are usually associated with focusing on artifacts or background.

D. CONVIT MODEL PERFORMANCE

ConViT model, which uses soft convolutional inductive biases as a structure of Vision Transformer, was tested in the context of colorectal cancer histopathology image classification. The model was trained on CrossEntropyLoss at a learning rate of 1e -4 with early stopping (patience=5) on the timm-pretrained convit -base with dropout rate = 0.15 using Adam as the optimizer and 20 epochs. ConViT on the CRC-VAL-HE-7K dataset (7,180 images in 9 classes) has a test accuracy of 88.52%, a test precision of 0.8935, a recall of 0.8852, and a F1 score of 0.8843, which is very good with data augmentation (random resized crop, horizontal flip). The training got converged in the early stopping threshold, denoting good generalization to imbalances between classes. Grad-CAM visualizations on the second-to-last transformer block revealed the model focusing on the most important pathological features in the tumor nuclei, and more local features of the image are captured because of its convolutional biases.

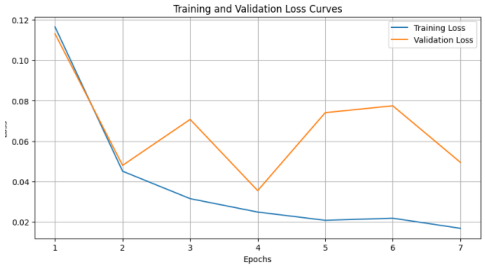

FIGURE 43: Loss Curve

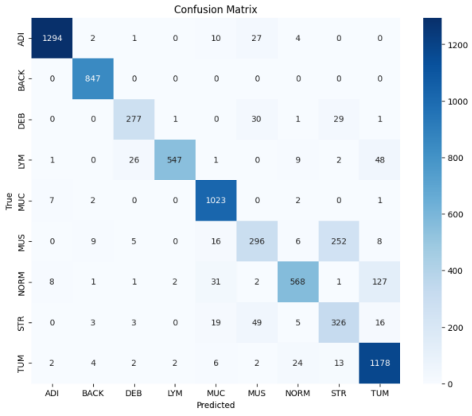

FIGURE 44: Confusion Matrix

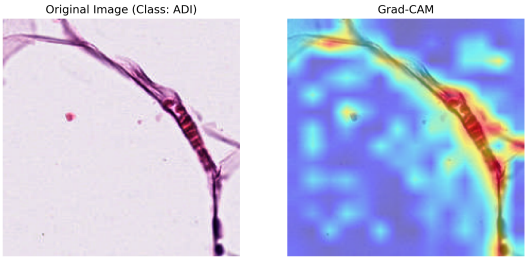

FIGURE 45: GRAD CAM Visualization of ADI

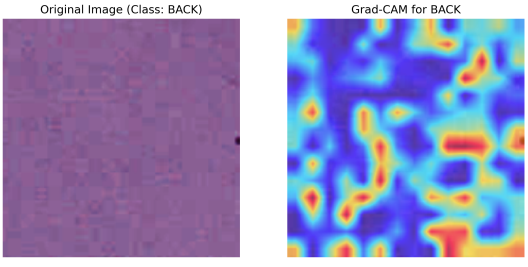

FIGURE 46: GRAD CAM Visualization of BACK

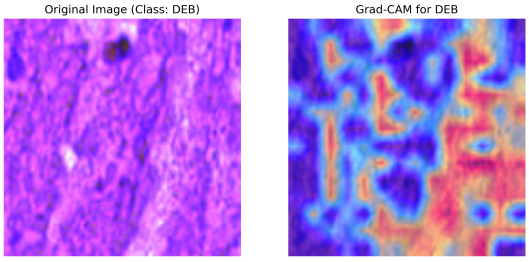

FIGURE 47: GRAD CAM Visualization of DEB

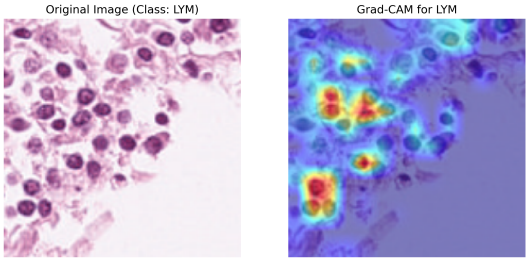

FIGURE 48: GRAD CAM Visualization of LYM

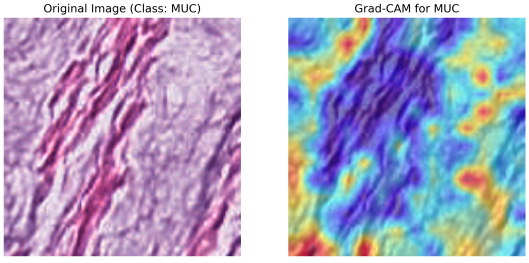

FIGURE 49: GRAD CAM Visualization of MUC

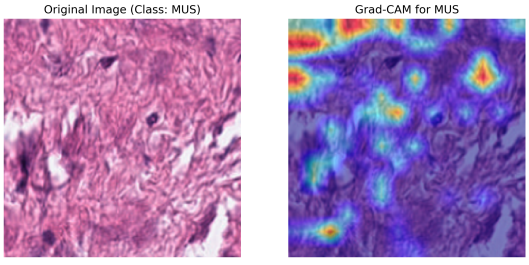

FIGURE 50: GRAD CAM Visualization of MUS

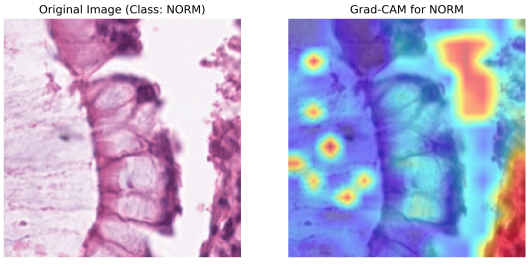

FIGURE 51: GRAD CAM Visualization of NORM

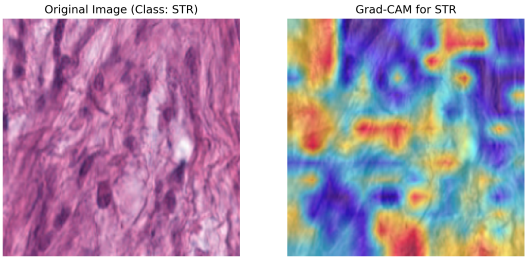

FIGURE 52: GRAD CAM Visualization of STR

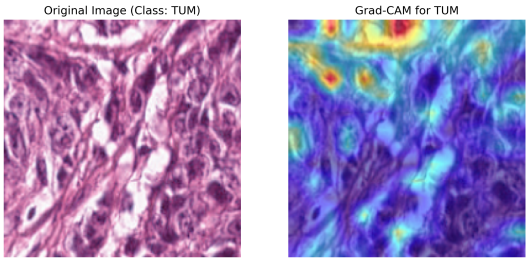

FIGURE 53: GRAD CAM Visualization of TUM

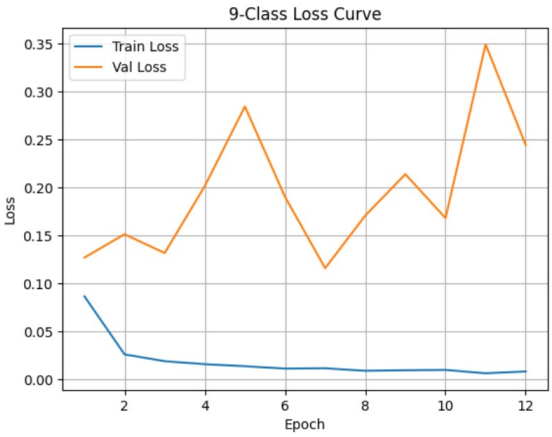

FIGURE 54: Loss Curve

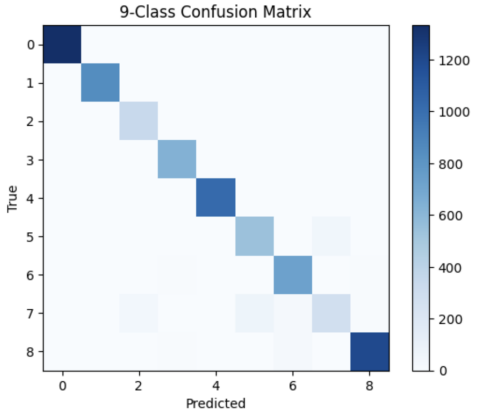

FIGURE 55: Confusion Matrix

E. SWIN TRANSFORMER MODEL PERFORMANCE

The Swin Transformer model was a hierarchical Vision Transformer, which used shifted window attention to classify histopathological colorectal cancer images on the CRC-VAL-HE-7K dataset. The model was trained on up to 100 epochs with early stopping (patience=5, min_epochs=12) on the AdamW optimizer and CrossEntropyLoss with a learning rate of 1e-4 using the pretrained swin_t of the torchvision. Swin Transformer reaches a test accuracy of 96.25% on the CRC-VAL-HE-7K dataset(7,180 images, 9 classes) and converges quickly, as indicated by loss curves. Precision (0.9621), recall (0.9625), and F1-score (0.9612) on classes on the CRC-VAL-HE-7K are reported in the classification report, and the confusion matrix indicates high scores in tumor-related classes. Grad-CAM visualizations (targeting the final norm layer) show that the model is interested in multi-scale characteristics such as tissue boundaries and nuclei, and window-based attention is more efficient.

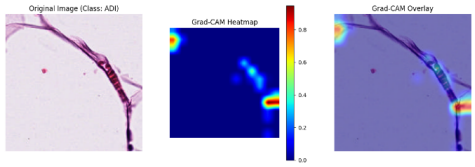

FIGURE 56: GRAD CAM Visualization of ADI

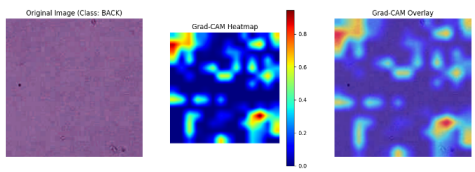

FIGURE 57: GRAD CAM Visualization of BACK

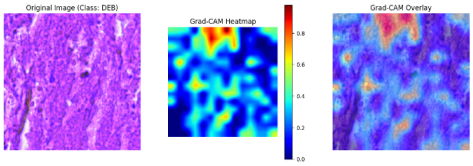

FIGURE 58: GRAD CAM Visualization of DEB

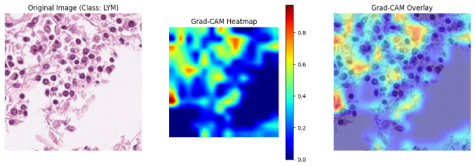

FIGURE 59: GRAD CAM Visualization of LYM

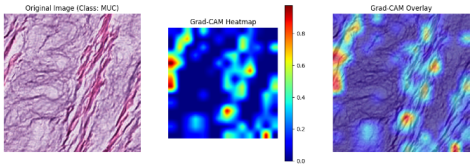

FIGURE 60: GRAD CAM Visualization of MUC

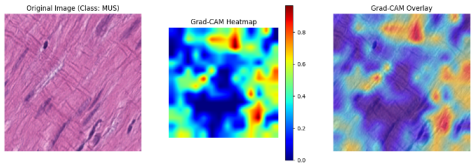

FIGURE 61: GRAD CAM Visualization of MUS

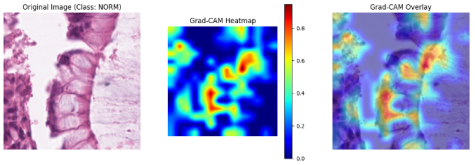

FIGURE 62: GRAD CAM Visualization of NORM

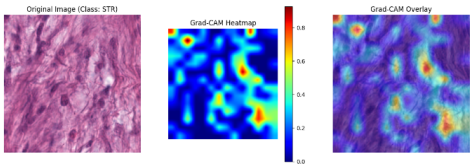

FIGURE 63: GRAD CAM Visualization of STR

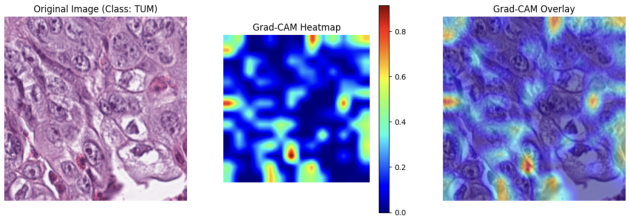

FIGURE 64: GRAD CAM Visualization of TUM

F. MODEL COMPARISON

Transformer model was the best model with a test accuracy of 96.25 on the CRC-VAL-HE-7K dataset, then EfficientNet-B0 with a test accuracy of 94.62% percent, EfficientViT-B0 with 92.00% percent, ViT with 93.68% percent and ConViT with 88.52% percent. ViT is well used as it has a global attention mechanism, but is computationally intensive (4.5 hours to train). Swin Transformer is efficient in multi-scale processing, which can be trained in about 3.5 hours, which is best suited to large pathology slides. ConViT offers hybrid advantages, but with less accuracy. All models were trained with data augmentation and a balanced dataset (5,000 images per class), which minimized the overfitting.

TABLE 2: Model Performance Comparison on CRC-VAL-HE-7K Test Set

Model	Accuracy	Precision	Recall	F1-Score
Vision Transformer(ViT)	0.9368	0.9504	0.9368	0.9348
EfficientViTB0	0.9200	0.9248	0.9214	0.9209
EfficientNetB0	0.9462	0.9465	0.9462	0.9445
ConViT	0.8852	0.8935	0.88.52	0.8843
Swin Transformer	0.9625	0.9884	0.9676	0.9779

V. CONCLUSION

This study evaluated the classification performance of different deep learning models for colorectal cancer histopathology images across multiple datasets. Among model variants tested, Swin Transformer demonstrated the highest accuracy, affirming the efficacy of hierarchical attention mechanisms for understanding multi-scale tissue patterns. At the same time, EfficientViT demonstrated the best efficiency-prediction performance tradeoff, and hence was best suited for deployment in resource-constrained clinical settings. These results demonstrate the potential of transformer-based and hybrid models to enhance computer-aided diagnosis for supporting pathologists in optimizing diagnostic accuracy while reducing workload.

Subsequent research will investigate enhancing model robustness by cross-institution generalization, employing stain normalization techniques to reduce variance between datasets, and multimodal fusion of information such as genomic and clinical data to develop more comprehensive decision-support systems for diagnosing colorectal cancer.

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